

Glossary

A

Acicular

Needle-like; the crystal habit of some minerals.

Adamantine

A bright, diamond-like lustre.

Adularescence

See *Opalescence*.

Allochromatic

Gems coloured by impurities, without which they would be colourless.

Asterism

A four- or six-ray star effect displayed by certain gems, including some sapphires and rubies, that have been cut *en cabochon*; the optical effect is caused by the reflection of light on fibrous or rutile inclusions.

B

Bezel

The part of the mounting that surrounds the girdle of a stone with a metal band.

Birefringence

In doubly refractive gems, this is the difference between the highest and lowest refractive indices. See also *Double refraction*.

Brilliant cut

A round cut featuring mathematically calculated proportions of triangular facets top and bottom, which are designed to maximize a diamond's fire and brilliance.

C

Cabochon

A polished cut with a domed upper surface and a flat or domed under surface; gems cut in this way are described as being cut *en cabochon*.

Cameo

A low-relief design that has been cut into layered stone or shell, with the background material cut away.

Carat

A unit of gemstone weight. One carat equals 0.2g (0.007oz). (Not to be confused with karat, a measure of gold purity.) See also *Karat*.

Chatoyancy, chatoyant

The cat's-eye effect shown on certain gems that have been cut *en cabochon*.

Clast

A fragment or grain of rock, usually broken off as a result of physical weathering; clastic rocks are a form of sedimentary rock composed of such clasts.

Cleavage

The way that some minerals break along planes determined by their atomic structure.

Crown

The top part of a cut stone, above the girdle.

Cryptocrystalline

An extremely fine-grained crystalline mineral habit, in which individual crystallized components can only be seen under a microscope.

Crystal

A solid with an ordered internal atomic structure that produces a typical external shape, along with characteristic physical and optical properties.

Crystal structure

The internal atomic structure of a crystal. All crystalline gems may be classified according to the symmetry of their structure: cubic; tetragonal; hexagonal; trigonal; orthorhombic; monoclinic; and triclinic.

Culet

The lowest part of a cut stone, either a point or a ridge.

Cushion

A square cut with rounded sides and corners.

Cut

The shaping of a gemstone by grinding and polishing; the shape of the final gem, as, for example, in brilliant cut.

D

Dendritic

Tree-like; the crystal habit exhibited by some minerals.

Diffraction

The splitting of white light into its constituent colours – the colours of the rainbow – when it passes through a hole or grating; the bending of light rays around the edge of an obstacle.

Dispersion

The splitting of white light into its constituent colours – the colours of the rainbow – as it passes through an inclined surface such as those on a prism or a faceted gem. Dispersion in gems is known as fire.

Double refraction (DR)

The splitting of light into two separate rays as it enters a gem. Each ray travels at a different speed and has its own refractive index.

E

Extrusive

A type of rock formed from lava that has either flowed onto the Earth's surface or was ejected from a volcanic vent.